

STUDENT'S NAME: _____

TEACHER'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

Year 12

MATHEMATICS ADVANCED ASSESSMENT

HALF-YEARLY

March 2008

Time allowed – 3 hours + 5 minutes reading time

DIRECTIONS TO CANDIDATES:

- Start each question on a new page.
- Show all relevant working.
- Use black or blue pen.
- **NO** liquid paper is to be used.
- Approved Maths aids and calculators may be used.

QUESTION 1 [12 marks]**Marks**

- (a) Evaluate, correct to three significant figures

2

$$1 - \sqrt{\frac{(2.044)^3}{34.5 - 1.2^2}}$$

- (b) Factorise $x^3 - 5x^2 + 6x$

2

- (c) A function $f(x)$ is defined as $f(x) = \begin{cases} |x-1| & \text{for } x \leq 0 \\ 4-x^2 & \text{for } x > 0 \end{cases}$

2

Evaluate $f(2) + f(0)$

- (d) Evaluate $\sum_{r=1}^3 r^2$

2

- (e) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

2

- (f) Evaluate $\int_0^3 e^{2x} dx$ in terms of e

2

QUESTION 2 [12 marks]

- (a) Find $\int \frac{3x^2}{x^3 + 5} dx$

2

- (b) Differentiate

i) $x^2 \ln 3x$

2

ii) $\frac{x}{3x-2}$

2

- (c) Given $\log_a 2 = 0.36$ and $\log_a 5 = 0.83$,
Show working to find the value of

(i) $\log_a 10$

2

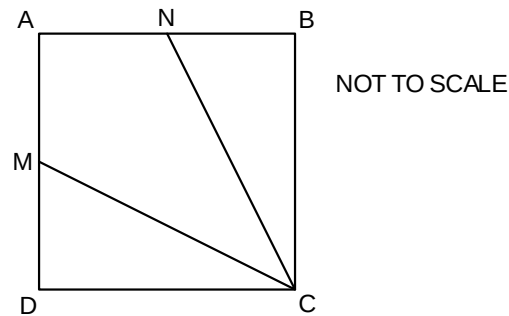
(ii) $\log_2 5$

2

- (d) By rationalising the denominators,

2

express $\frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}}$ in simplest form

QUESTION 3 [12 marks]**Marks****(a)**

ABCD is a square. Lines are drawn from C to M and N , the midpoints of AD and AB respectively.

(i) Show that $\triangle CBN \equiv \triangle CDM$ **3**

(ii) Prove that $MC = NC$ **1**

(b) Let T_n = the n th term of a series, and S_n = the sum of n terms of a series. For a particular arithmetic series, $S_n = n(2n - 1)$

(i) Find S_1 and S_2 **2**

(ii) Find the expression for T_n for this series **2**

(c) For the curve $y = \frac{1}{x+1} + 4$

(i) State the domain and range **2**

(ii) Sketch the graph showing all important features **2**

QUESTION 4 [12 marks]**Marks**

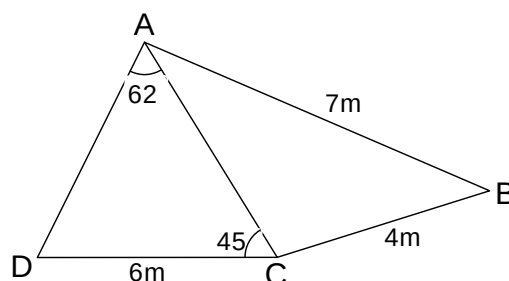
(a) For the following diagram, find

(i) find AC

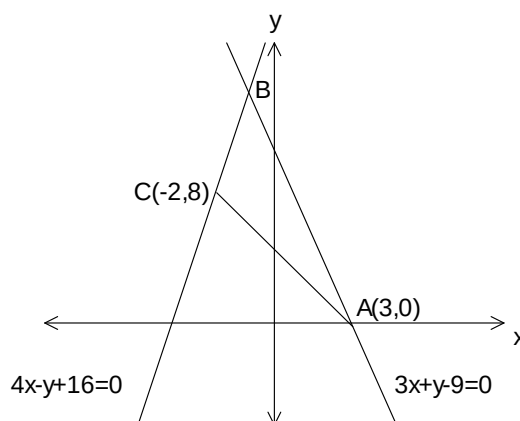
2

(ii) find the size of $\angle ABC$ (correct to the nearest degree)

2



(b) In the diagram below the lines $3x + y - 9 = 0$ and $4x - y + 16 = 0$ intersect at the point B. The point A has the coordinates $A(3,0)$ and the point C has the coordinates $C(-2,8)$.



(i) Show that the line AC has the equation $8x + 5y - 24 = 0$

2

(ii) B is the intersection point of the lines $3x + y - 9 = 0$ and $4x - y + 16 = 0$, show that the co-ordinates of B is $(-1,12)$

1

(iii) Find the length of AC. Leave your answer as a surd.

2

(iv) Find the perpendicular distance from point B to AC. Leave your answer as a surd.

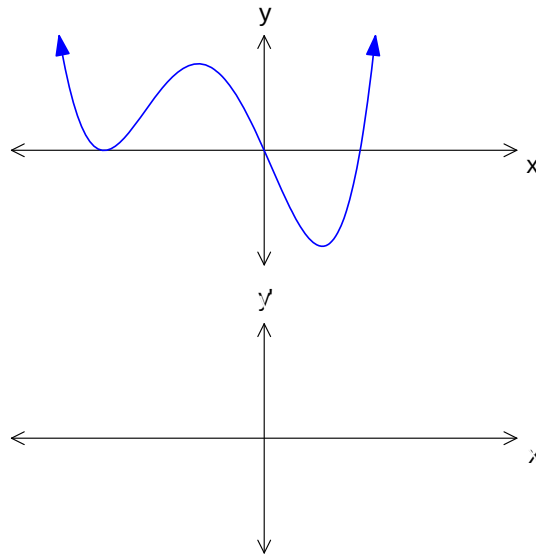
2

(v) Hence find the area of $\triangle ABC$

1

QUESTION 5 [12 marks]**Marks**

- (a) Copy the diagram and below the axis sketch the derivative function

1

- (b) Consider the function $f(x) = 3x^4 - 4x^3$
- (i) Show that $f'(x) = 12x^2(x-1)$ **1**
- (ii) Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. **4**
- (iii) Sketch the graph of the curve $y = f(x)$, showing stationary points, **1**
- (iv) Find the coordinates of any points of inflection. **2**
- (v) Find the values of x for which the graph of $y = f(x)$ is concave down. **2**
- (vi) Find the greatest values of $y = f(x)$ in the domain $-1 \leq x \leq 2$ **1**

QUESTION 6 [12 marks]**Marks**

(a) If α , β are roots of the quadratic equation $6x^2 - x + 5 = 0$, find:

(i) $\alpha + \beta$ 1

(ii) $\alpha \times \beta$ 1

(iii) $\alpha^2 + \beta^2$ 2

(iv) $\alpha^3 + \beta^3$ 2

(b) Given the parabola $y = x^2 + 6x + 7$, find:

(i) focal length 1

(ii) vertex 1

(iii) Find the equation of the normal at $x = -1$, in general form. 2

(c) Solve for $2\cos\theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$ 2

QUESTION 7 [12 marks]

(a) Show that $\frac{3x+3}{x^2-9} = \frac{1}{x+3} + \frac{2}{x-3}$ 2

Hence find $\int \frac{3x+3}{x^2-9} dx$.

(b) Find the volume of the solid of revolution formed by rotating 4
the curve $y = x - \frac{1}{x}$ about the x-axis between $x = 1$ and $x = 4$
Answer in terms of π .

(c) Find $\frac{d}{dx}(x^2 - 3)^5$ and 2
Hence find $\int x(x^2 - 3)^4 dx$

(d) Find the largest term less than 10 000 in the series $1 + 4 + 7 + 10$ 2

(e) Find the values of K for which the equation 2
 $x^2 - Kx + 4 = 0$, has two real distinct roots

QUESTION 8 [12 marks]**Marks**

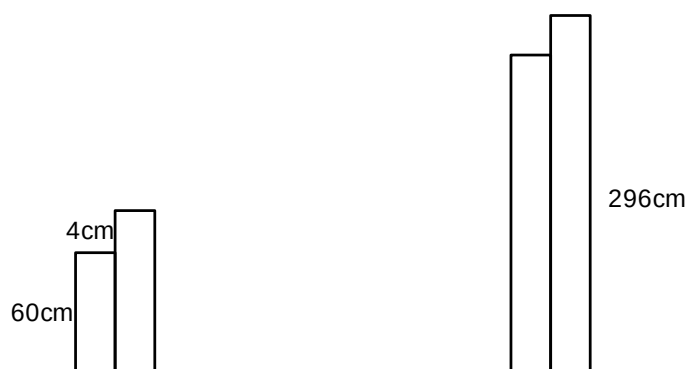
- (a) Differentiate:
- (i) 4^x 1
- (ii) $\log_{10} x$ 2
- (b) Solve the equation $2 \ln x = \ln(6 - x)$ 3
- (c) By expressing $0.\overline{25}$ as the sum of an infinite geometric series, find the simple equivalent fraction for $0.\overline{25}$. 2
- (d) The gradient of the curve $y = f(x)$ is given by $f'(x) = (2x + 1)^5$. Find the equation of the curve if it passes through the point (1, 5) 2
- (e) Find the values A , B and C if $x^2 + 5 \equiv A(x - 1)^2 + B(x - 1) + C$ 2

QUESTION 9 [12 marks]

- (a) Solve for x : $3^{2x} - 4(3^x) + 3 = 0$ 2
- (b) (i) Sketch the graph $y = |2x| - 4$ for $-2 \leq x \leq 4$. Showing all features of the graph. 2
- (ii) Hence or otherwise, evaluate $\int_{-2}^4 |2x| - 4 \, dx$ 2
- (c) A farmer is building a wheat silo in the shape of a closed cylinder of radius r metres. The silo is to be made from galvanised iron sheeting and is to have a capacity of 300m^3 .
- (i) Find an expression for the height of the silo in terms of r . 1
- (ii) Show that the surface area A , of the silo is given by the equation $A = 2\pi r^2 + \frac{600}{r}$ 2
- (iii) Hence find the minimum area of galvanised iron sheeting needed to make the silo, leaving your answer in one decimal place. 3

QUESTION 10 [12 marks]**Marks**

- (a) Find the area enclosed between the curve $y = x^2$ and the line $y = x + 2$ **3**
- (b) (i) Shade the region bounded by the curve $y = \log_e x$, the y -axis, the x -axis and the line $y = 2$. **1**
- (ii) Calculate the volume of the solid of revolution formed when this region is rotated about the y -axis, in terms of π . **3**
- (c) The following diagram represents a retaining wall in a nursery made from timber. Each length of the timber is 4cm longer than the proceeding one. The last length of timber is 296cm long and the first is 60cm long.



Find:

- (i) How many lengths of timber make up the wall **2**
- (ii) The total length of the timber needed to make the retaining wall. **2**
- (iii) If each of timber is 10cm wide, find the area of the wall in square metres **1**

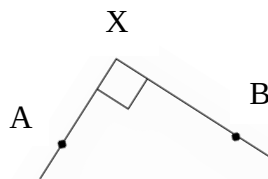
End of Paper

The third term and the tenth term of an arithmetic series are 10 and 31 respectively. Find the:

- i) first term and the common difference 2
- ii) sum of the first ten terms of the series 1
- (c) Using Simpson's Rule with three function values, find an approximate value for the area represented by the definite integral 3
- $$\int_2^3 \cos^2 x \, dx$$
- (d) For what values of k does the quadratic equation 1
- $$kx^2 + kx + 1 = 0$$
- have no real roots?

Find the equation of the tangent to the curve $y = \ln(x^2 + 2)$ 4
at the point where $x = 1$. Answer in general form

- (b)
(c)



Solve $\log_{27} 32 = x \log_3 2$ without the aid of a calculator. 2
Show all working

- (f)

a) 0.49175 ✓
 $= 0.492$ (3 sig fig) ✓

b) $x^3 - 5x^2 + 6x$
 $= x(x^2 - 5x + 6)$ ✓
 $= x(x-3)(x-2)$ ✓

c) $f(2) = 4 - 2^2 = 0$
 $f(0) = 10 - 1 = 9$ ✓

$f(2) + f(0) = 0 + 9 = 9$ ✓

d) $\sum_{r=1}^3 r^2 = 1^2 + 2^2 + 3^2$ ✓
 $= 1 + 4 + 9$
 $= 14$ ✓

e) $\lim_{x \rightarrow 3} \frac{(x+3)(x-3)}{(x-3)} = \lim_{x \rightarrow 3} x+3$ ✓

f) $\int_0^3 e^{2x} dx = \left(\frac{1}{2} e^{2x} \right)_0^3$ ✓
 $= \frac{1}{2} e^6 - \frac{1}{2} e^0$ ✓
 $= \frac{1}{2} e^6 - \frac{1}{2}$ ✓

Q2 a) $\int \frac{3x^2}{x^3+5} dx = \ln(x^3+5) + C$ ✓

b) (i) $y = x^2 \ln 3x$

$u = x^2$ $v = \ln 3x$

$u' = 2x$ $v' = \frac{1}{3x} = \frac{1}{x}$ ✓

$y' = 2x \ln 3x + \frac{1}{x} \times x^2$ ✓
 $= 2x \ln 3x + x$ ✓

(ii) $y = \frac{x}{3x-2}$ $u = x$ $v = 3x-2$

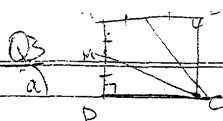
$u' = 1$ $v' = 3$ ✓

$y' = \frac{1 \times (3x-2) - 3x}{(3x-2)^2}$
 $= \frac{-2}{(3x-2)^2}$ ✓

c) i) $\log_a 10 = \log_a 2 + \log_a 5$ ✓
 $= 0.36 + 0.83$
 $= 1.19$ ✓

(ii) $\log_2 5 = \frac{\log_2 5}{\log_2 2} = \frac{0.83}{0.36}$ ✓
 $= 2.31$ ✓

d) $\frac{1}{(3-\sqrt{2})(3+\sqrt{2})} + \frac{1}{(3+\sqrt{2})(3-\sqrt{2})} = \frac{6}{9-2}$ ✓
 $= \frac{6}{7}$ ✓



DM = AM = AN = NB (M and N bisect equal sides of square)

In $\triangle CBN$ and $\triangle CDM$.

BC = DC (equal sides of square) ✓

NB = MD (shown above) ✓

$\angle MDC = \angle NBC$ (ls of a square is 90°) ✓

$\therefore \triangle CBN \cong \triangle CDM$ (SAS) ✓

ii) $MC = NC$ (Sides of congruent $\triangle$ s are =) ✓

b) i) $S_1 = 1(2 \times 1 - 1) = 1$ ✓ $S_2 = 2(2 \times 2 - 1) = 6$ ✓

(ii) $T_n = S_n - S_{n-1}$

$S_{n-1} = (n-1)(2(n-1) - 1)$

$= (n-1)(2n-3)$

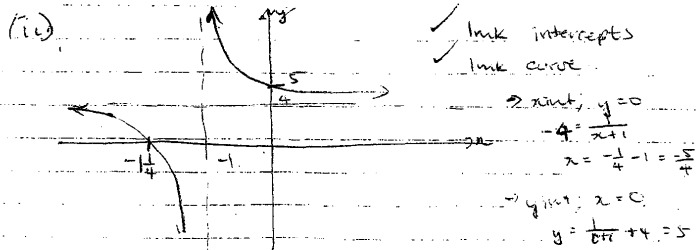
$= 2n^2 - 5n + 3$ ✓

$T_n = (2n^2 - n) - (2n^2 - 5n + 3)$ ✓

$= 4n - 3$

c) i) Domain: $\{ \text{all real } x, \text{ except } x \neq -1 \}$ ✓

Range: $\{ \text{all real } y, \text{ except } y \neq 4 \}$ ✓



Q4 a) i) $\frac{AC}{\sin 73^\circ} = \frac{6}{\sin 62^\circ}$

$AC = 6.49...$ ✓

ii) $c^2 = a^2 + b^2 - 2ab \cos C$

$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

$\cos C = \frac{7^2 + 4^2 - AC^2}{2 \times 7 \times 4}$

$C = 66^\circ$ ✓

(nearest degree)

b) i) $m = \frac{8-0}{-2-3} = -\frac{8}{5}$ ✓

at (3,0)

$y - 0 = -\frac{8}{5}(x - 3)$

$-5y = 8x - 24$

$8x + 5y - 24 = 0$ ✓

ii) Sub (-1, 12)

① $3(-1) + 12 - 9 = 0$ ✓ lies on line

② $4(-1) - (12) + 14 = 0$ ✓ lies on line

$B(-1, 12)$ is the pt of intersection ✓

iii) $AC = \sqrt{(3-2)^2 + (0)^2} = \sqrt{1} = 1$ ✓

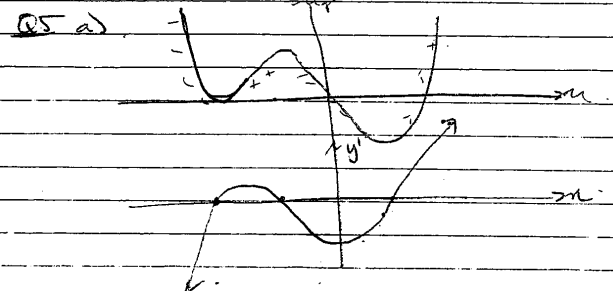
iv) $d = \frac{|8(-1) + 5(12) - 24|}{\sqrt{8^2 + 5^2}} = \frac{28}{\sqrt{89}}$ ✓

$$(u) A = \frac{1}{2}bh$$

$$= \frac{1}{2} \times \sqrt{84} \times \frac{28}{\sqrt{49}}$$

$$= 14 \text{ units}^2$$

✓



$$b) f(x) = 3x^4 - 4x^3$$

$$(i) f(x) = 12x^3 - 12x^2$$

$$= 12x^2(x-1)$$

$$(ii) \text{ stat pt when } f'(x) = 0$$

$$0 = 12x^2(x-1)$$

$$x = 0 \text{ or } x = 1$$

$$f(0) = 0 \text{ or } f(1) = -1$$

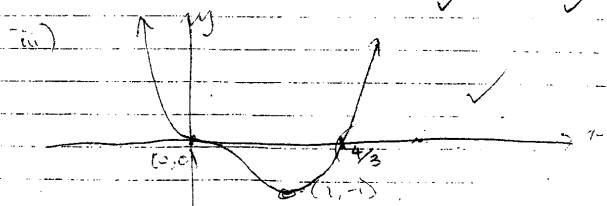
$$f''(x) = 36x^2 - 24x$$

$$f''(0) = 0$$

$$f''(1) = 12 > 0$$

$(0,0)$ is a horizontal inflection

$(1,-1)$ is a min pt



$$Q5 iv) f''(x) = 36x^2 - 24x$$

$$0 = 36x^2 - 24x$$

$$0 = x(3x-2)$$

$$x = 0$$

$$x = \frac{2}{3}$$

$$f(0) = 0$$

$$f(\frac{2}{3}) =$$

pts of inflection are $(0,0)$ and $(\frac{2}{3}, -\frac{16}{27})$

v) concave down when $f''(x) < 0$

$$x(3x-2) < 0$$

$$0 < x < \frac{2}{3}$$

$\therefore$ concave down when $0 < x < \frac{2}{3}$

$$vi) f(-1) = 3(-1)^4 - 4(-1)^3$$

$$= 7$$

$$f(2) = 3(2)^4 - 4(2)^3$$

$$= 16$$

$\therefore$ max value is 16

$$Q6 a) 6x^2 - x + 5 = 0$$

$$(i) x + p = \frac{1}{6}$$

$$(ii) x p = \frac{5}{6}$$

$$(iii) x^2 + p^2 = x^2 + p^2 + 2xp - 2xp$$

$$= (x+p)^2 - 2xp$$

$$= \left(\frac{1}{6}\right)^2 - 2\left(\frac{5}{6}\right)$$

$$= -\frac{59}{36}$$

$$(iv) x^3 + p^3 = (x+p)(x^2 - xp + p^2)$$

$$= \left(\frac{1}{6}\right)\left(\frac{1}{36} - \frac{5}{6}\right)$$

$$= -\frac{89}{216}$$

(26)

$$b) y = x^2 + 6x + 7 =$$

$$y = x^2 + 6x + 9 - 9 + 7$$

$$y = (x+3)^2 + 7$$

$$(x+3)^2 = (y-7)$$

$$(i) \text{ focal length} = \frac{1}{4} \quad \checkmark$$

$$(ii) \text{ vertex } (-3, -2) \quad \checkmark$$

$$(iii) y = 2x + 6$$

$$\text{at } x = 1 \quad m_N = 4$$

$$y = 1 - 6 + 7 = 2$$

$$\text{at } (-4, 2)$$

$$m_N = \frac{1}{4} \quad \checkmark$$

$$y - 2 = \frac{1}{4}(x + 1)$$

$$x - 4y + 7 = 0 \quad \checkmark$$

$$c) \text{ work} = \frac{1}{2} \quad \checkmark \quad \frac{3}{T} \quad \checkmark$$

$$P = 60 \text{ or } 300 \quad \checkmark$$

$$(27) a) \text{ RHS} = \frac{1}{x+3} + \frac{2}{x-3}$$

$$= \frac{x-3 + 2(x+3)}{(x+3)(x-3)}$$

$$= \frac{3x+3}{x^2-9}$$

$$= \text{LHS} \quad \checkmark$$

$$\int \frac{3x+3}{x^2-9} dx = \int \frac{1}{x+3} + \frac{2}{x-3} dx$$

$$= \ln(x+3) + 2 \ln(x-3) \quad \checkmark + C$$

$$= \ln(x+3)(x-3)^2 + C$$

$$(b) V = \pi \int_1^4 y^2 dx$$

$$y = x - \frac{1}{x}$$

$$y^2 = (x - \frac{1}{x})^2$$

$$= x^2 - 2x \cdot \frac{1}{x} + \frac{1}{x^2}$$

$$= x^2 - 2 + \frac{1}{x^2} \quad \checkmark$$

$$V = \pi \int_1^4 (x^2 - 2 + \frac{1}{x^2}) dx \quad \checkmark$$

$$= \pi \left[\frac{x^3}{3} - 2x - \frac{1}{x} \right]_1^4 \quad \checkmark$$

$$= \pi \left[\left(\frac{64}{3} - 8 - \frac{1}{4} \right) - \left(\frac{1}{3} - 2 - 1 \right) \right]$$

$$= \frac{65\pi}{4} \quad \checkmark$$

$$c) \frac{d}{dx} (x^2-3)^5 = 5(x^2-3)^4 \cdot 2x = 10x(x^2-3)^4 \quad \checkmark$$

$$\frac{1}{15} \int 3x(x^2-3)^5 dx = \frac{1}{15} (x^2-3)^5 + C \quad \checkmark$$

$$d) AP \quad a=1 \quad d=3$$

$$T_n = a + (n-1)d < 10000 \quad \checkmark$$

$$1 + (n-1)3 < 10000$$

$$n < \frac{9999}{3}$$

$$n < 3333$$

$$n = 3333$$

$$\therefore \text{largest term} = T_{3333} = 9997 \quad \checkmark$$

$$e) x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$2 \text{ distinct roots if } \Delta > 0 \quad \checkmark$$

$$\Delta = b^2 - 4ac > 0$$

$$k^2 - 4 \times 1 \times 4 > 0$$

$$k^2 - 16 > 0$$

$$(k+4)(k-4) > 0$$

$$\text{Distinct roots} \Rightarrow k < -4 \text{ or } k > 4 \quad \checkmark$$

$$Q8) a) y = 4^x$$

$$y = \ln 4 \cdot 4^x \quad \checkmark$$

$$b) y = \log_{10} x = \frac{\log x}{\log 10} \quad \checkmark$$

$$y = \frac{1}{\log 10} \times \frac{1}{x}$$

$$= \frac{1}{x \log 10} \quad \checkmark$$

$$b) \ln x^2 = \ln(6-x) \quad \checkmark$$

$$x^2 = 6-x$$

$$x^2 + x - 6 = 0$$

$$(x+3)(x-2) = 0$$

$$x = -3, 2 \quad \checkmark$$

$$\text{But } x \neq -3 \therefore x = 2$$

$$e) 0.25 = 0.25 + 0.0025 + 0.000025 + \dots \quad \checkmark$$

$$GP \quad a = 0.25 \quad r = 0.001$$

$$S_{\infty} = \frac{a}{1-r} = \frac{0.25}{1-0.001}$$

$$= \frac{0.25}{0.99}$$

$$= \frac{25}{99} \quad \checkmark$$

Q8 - (cont)

a) $f'(x) = (2x+1)^5$

$f(x) = \frac{(2x+1)^6}{6 \times 2} + C$ ✓

at (1,5)

$5 = \frac{(2(1)+1)^6}{12} + C$

$C = -55\frac{3}{4}$ ✓ $-55\frac{3}{4}$

$f(x) = \frac{(2x+1)^6}{12} - 55\frac{3}{4}$

c) $x^2 + 5 = A(x-1)^2 + B(x-1) + C$

$x^2 + 5 = A(x^2 - 2x + 1) + Bx - B + C$ ✓

$x^2 + 5 = Ax^2 - 2x + A + Bx - B + C$ ✓

$A = 1$

$-2A + B = 0$

$B = 2$

$A - B + C = 5$

$1 - 2 + C = 5$

$C = 6$

$A = 1, B = 2, C = 6$ ✓

Q9

a) $3^{2x} - 4(3^x) + 3 = 0$

$(3^x - 3)(3^x - 1) = 0$ ✓

$3^x = 3$ or $3^x = 1$

$x = 1$ or $x = 0$ ✓

b) $y = |2x| - 4$

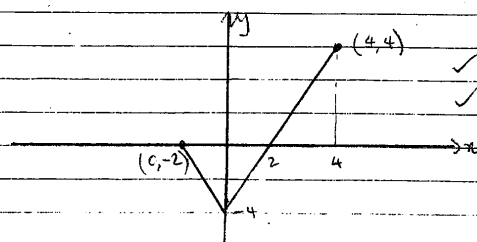
(i) y int; $x = 0$
 $y = -4$

x int $y = 0$

$|2x| = 4$

$x = \pm 2$

$x = 4, y = 4$



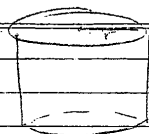
✓ 1 mk intercepts

✓ 1 mk domain

(ii) $\int_{-4}^4 |2x| - 4 \, dx = \int_{-2}^0 |2x| - 4 \, dx + \int_0^2 |2x| - 4 \, dx$
 $= \frac{1}{2} \times 2 \times 4 - \frac{1}{2} \times 4 \times 4$
 $= -4$ ✓

✓ 1 mk negative

Q98



$$V = 300$$

$$V = \pi r^2 h$$

$$300 = \pi r^2 h$$

$$h = \frac{300}{\pi r^2}$$

(i)

$$(ii) \quad A = 2\pi r^2 + 2\pi r h$$

$$A = 2\pi r^2 + 2\pi r \left(\frac{300}{\pi r^2} \right)$$

$$A = 2\pi r^2 + \frac{600}{r}$$

$$A = 2\pi r^2 + 600r^{-1}$$

$$(iii) \quad \frac{dA}{dr} = 4\pi r - \frac{600}{r^2}$$

$$\text{min when } \frac{dA}{dr} = 0$$

$$0 = 4\pi r - \frac{600}{r^2}$$

$$4\pi r^3 = 600$$

$$r = \sqrt[3]{\frac{600}{4\pi}} = 3.627$$

r	3	3.627	4
$\frac{dA}{dr}$	-28	0	12.76

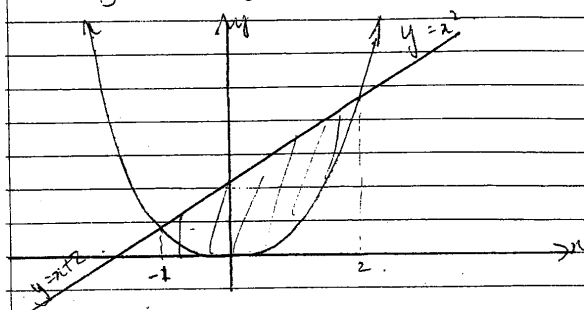
min value at 3.627

$$A = 2\pi \times 3.627^2 + \frac{600}{3.627}$$

$$A = 218.85 \text{ units}^2$$

Q10

$$a) \quad y = x^2 - 1 \quad y = x + 2 \quad (2)$$



$$x^2 = x + 2$$

$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$x = 2, x = -1$$

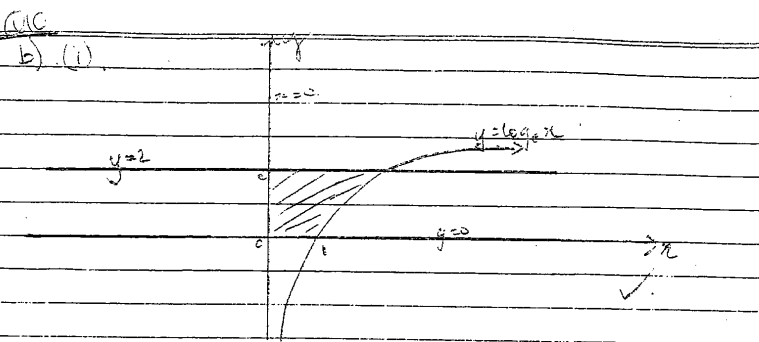
$$\int_{-1}^2 (x + 2) - x^2 \, dx =$$

$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2$$

$$= \left[\left(\frac{2^2}{2} + 2 \times 2 - \frac{2^3}{3} \right) - \left(\frac{(-1)^2}{2} + 2(-1) - \frac{(-1)^3}{3} \right) \right]$$

$$= \frac{10}{3} - \left(-\frac{7}{6} \right)$$

$$= \frac{9}{2} \text{ units}^2$$



(ii) $V = \pi \int x^2 dy$

$y = \log_e x$
 $x = e^y$
 $x^2 = e^{2y}$ ✓

$V = \pi \int_0^2 e^{2y} dy$ ✓

$V = \pi \left[\frac{1}{2} e^{2y} \right]_0^2$

$V = \frac{\pi}{2} [e^4 - e^0] = \frac{\pi}{2} (e^4 - 1)$

$V = \frac{\pi e^4 - \pi}{2}$ ✓

Q10. c) $60 + 64 + 68 + \dots + 296$

(i) AP: $a=60$ $d=4$ $T_n=296$ ✓

$= 296 = 60 + (n-1)4$
 $n=60$ ✓

∴ There are 60 lengths.

(ii) $S_n = \frac{n}{2} (a+l)$ ✓

$S_{60} = \frac{60}{2} (60 + 296)$

$S_{60} = 10680 \text{ cm}$ ✓

∴ 10680 cm of timber needed.

(iii) 106.80 m of timber

0.1 m wide

$A = 10.68 \text{ m}^2$ ✓

(if given 106800 cm^2 $\frac{1 \text{ m}^2}{10000}$ only)